

SHETH L.U.J. AND SIR M.V. COLLEGE

Aim: Understanding TinyOS Architecture and Execution Model

Code:

```
import time
import queue
import threading

class TinyOSSchedulerSimulation:
    def __init__(self):
        self.task_queue = queue.Queue(maxsize=8)
        self.is_running = True
        self.system_lock = threading.Lock()

    def post_task(self, task_name, computation_time):
        """Simulates the nesC 'post taskName()' command."""
        try:
            self.task_queue.put_nowait((task_name, computation_time))
            print(f"[SYSTEM]: Task '{task_name}' successfully posted to
FIFO queue.")
        except queue.Full:
            print(f"[WARNING]: FIFO Task Queue Full! Task '{task_name}'
dropped.")

    def trigger_hardware_event(self, event_name, bound_task_name,
comp_time):
        """Simulates an asynchronous hardware interrupt."""
        print(f"\n[HARDWARE INTERRUPT]: Event '{event_name}' fired!")

        self.post_task(bound_task_name, comp_time)

    def run_scheduler_loop(self):
        """Main TinyOS scheduler loop."""
        print("[INIT]: TinyOS Scheduler Loop Initialized. Waiting for
tasks...")

        while self.is_running or not self.task_queue.empty():
            try:
                task_name, comp_time = self.task_queue.get(timeout=1)
```

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```
print(f"\n[SCHEDULER]: Dispatching '{task_name}' (FIFO
Front)...")

    with self.system_lock:
        start_time = time.time()

        time.sleep(comp_time)

        elapsed_time = time.time() - start_time

        print(
            f"[TASK COMPLETE]: '{task_name}' completed in
{elapsed_time:.2f} seconds."
        )

        self.task_queue.task_done()

    except queue.Empty:
        if not self.is_running:
            break

        print("[POWER]: No pending tasks. MCU entering low-power
sleep mode...")
        time.sleep(0.5)

if __name__ == "__main__":

    simulation = TinyOSSchedulerSimulation()

    scheduler_thread =
threading.Thread(target=simulation.run_scheduler_loop)
    scheduler_thread.start()

    time.sleep(0.5)

    simulation.trigger_hardware_event(
        "Timer0.fired",
```

```
        "ReadSensorTask",
        1.5
    )

    simulation.trigger_hardware_event(
        "Radio.receive",
        "ProcessPacketTask",
        0.8
    )

    time.sleep(0.4)

    simulation.trigger_hardware_event(
        "ADC.dataReady",
        "FilterDataTask",
        0.5
    )

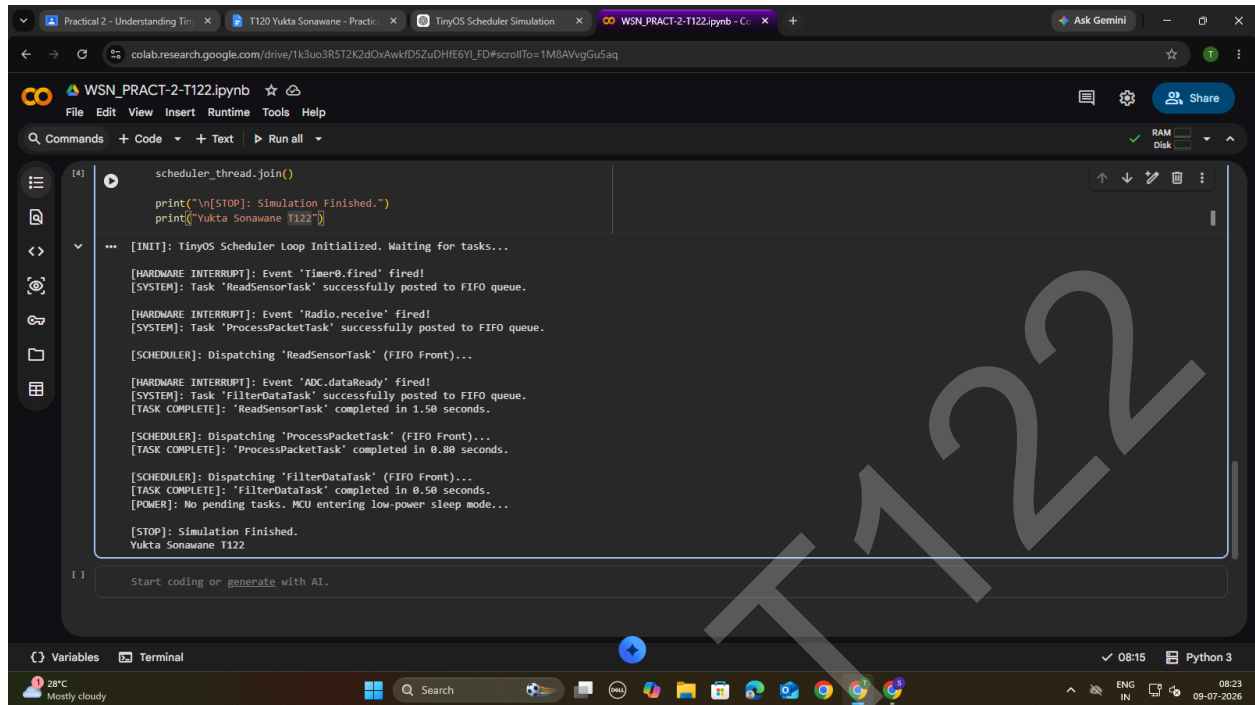
    time.sleep(4)

    simulation.is_running = False
    scheduler_thread.join()

    print("\n[STOP]: Simulation Finished.")
    print("Yukta Sonawane T122")
```

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OUTPUT:



The screenshot displays a Jupyter Notebook interface with a dark theme. The notebook is titled "WSN_PRACT-2-T122.ipynb" and is open in a web browser. The output area shows the execution of a simulation. The code cell contains a single line: `scheduler_thread.join()`. The output area displays the following text:

```
[4]: scheduler_thread.join()

print("\n[STOP]: Simulation Finished.")
print("Yukta Sonawane T122")

[INIT]: TinyOS Scheduler Loop Initialized. Waiting for tasks...

[HARDWARE INTERRUPT]: Event 'Timer0.fired' fired!
[SYSTEM]: Task 'ReadSensorTask' successfully posted to FIFO queue.

[HARDWARE INTERRUPT]: Event 'Radio.receive' fired!
[SYSTEM]: Task 'ProcessPacketTask' successfully posted to FIFO queue.

[SCHEDULER]: Dispatching 'ReadSensorTask' (FIFO Front)...

[HARDWARE INTERRUPT]: Event 'ADC.dataReady' fired!
[SYSTEM]: Task 'FilterDataTask' successfully posted to FIFO queue.
[TASK COMPLETE]: 'ReadSensorTask' completed in 1.50 seconds.

[SCHEDULER]: Dispatching 'ProcessPacketTask' (FIFO Front)...
[TASK COMPLETE]: 'ProcessPacketTask' completed in 0.80 seconds.

[SCHEDULER]: Dispatching 'FilterDataTask' (FIFO Front)...
[TASK COMPLETE]: 'FilterDataTask' completed in 0.50 seconds.
[POWER]: No pending tasks. MCU entering low-power sleep mode...

[STOP]: Simulation Finished.
Yukta Sonawane T122
```

The bottom of the notebook shows a status bar with "Variables", "Terminal", and "Python 3". The system tray at the bottom of the screen shows the date and time as "09-07-2026" and "08:15".

YUKTA SONAWANE T122
WSN PRACTICAL NO : 2